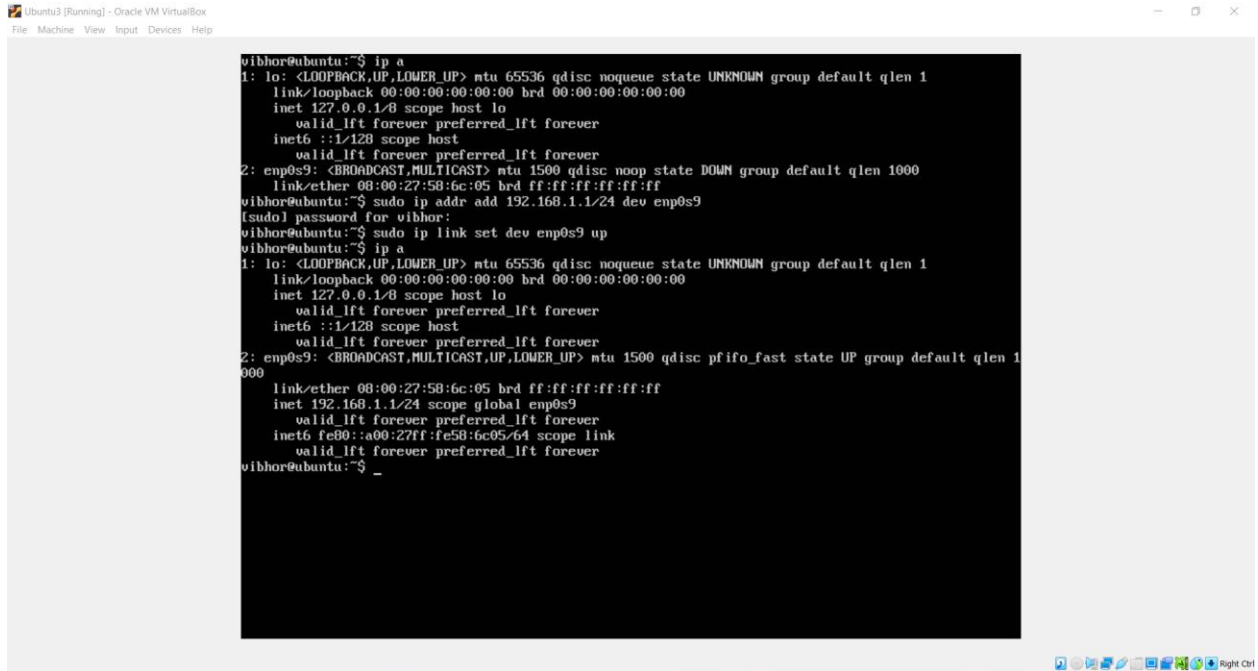


Exercise 1

Task 1

Assign private IP addresses to each interface

For interface iface1(192.168.1.1) of VM1



```
vibhor@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:58:6c:05 brd ff:ff:ff:ff:ff:ff
vibhor@ubuntu:~$ sudo ip addr add 192.168.1.1/24 dev enp0s9
[sudo] password for vibhor:
vibhor@ubuntu:~$ sudo ip link set dev enp0s9 up
vibhor@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:58:6c:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 scope global enp0s9
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe58:6c05/64 scope link
        valid_lft forever preferred_lft forever
vibhor@ubuntu:~$ _
```

For interfaces iface1(192.168.1.2) and iface2(192.168.2.1) of VM2

```
Ubuntu2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

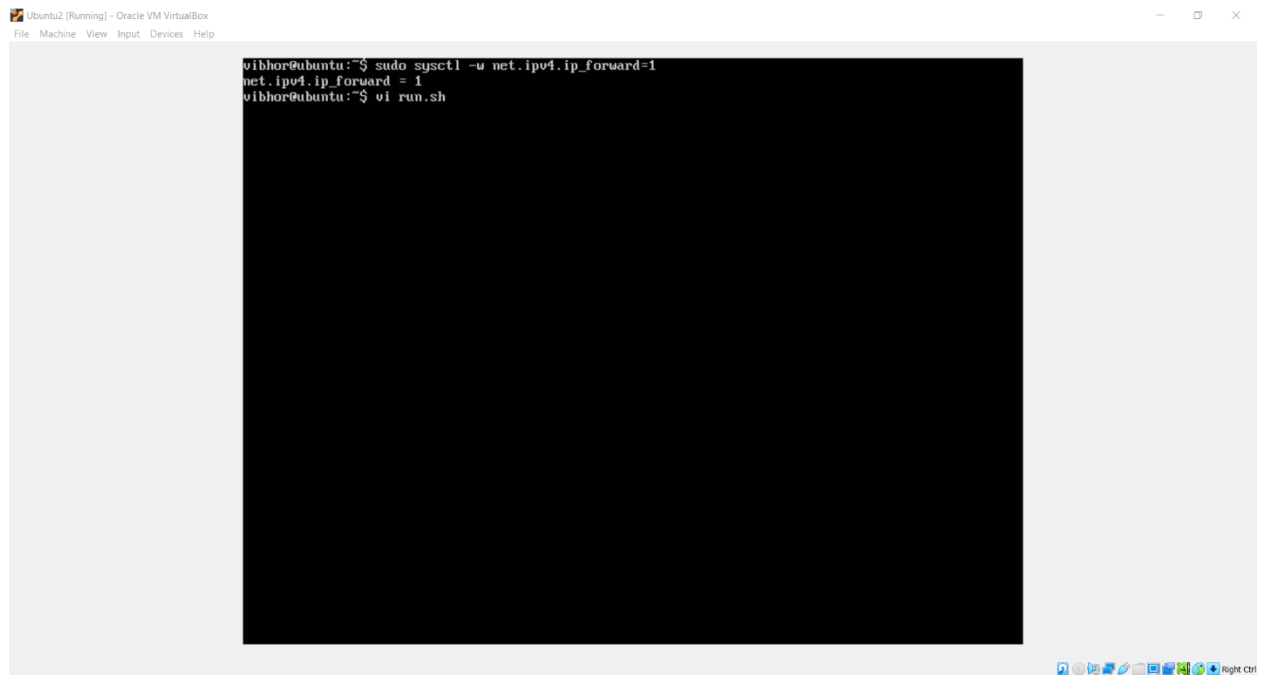
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:fa:60:4b brd ff:ff:ff:ff:ff:ff
inet6 fe80::a00:27ff:fe6a:604b/64 scope link
valid_lft forever preferred_lft forever
3: enp0s10: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
link/ether 08:00:27:0b:fe:ba brd ff:ff:ff:ff:ff:ff
vibhor@ubuntu:~$ sudo ip addr add 192.168.1.2/24 dev enp0s9
[sudo] password for vibhor:
vibhor@ubuntu:~$ sudo ip link set dev enp0s9 up
vibhor@ubuntu:~$ sudo ip addr add 192.168.2.1/24 dev enp0s10
vibhor@ubuntu:~$ sudo ip link set dev enp0s10 up
vibhor@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:fa:60:4b brd ff:ff:ff:ff:ff:ff
inet 192.168.1.2/24 scope global enp0s9
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe6a:604b/64 scope link
valid_lft forever preferred_lft forever
3: enp0s10: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:0b:fe:ba brd ff:ff:ff:ff:ff:ff
inet 192.168.2.1/24 scope global enp0s10
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe0b:feba/64 scope link
valid_lft forever preferred_lft forever
vibhor@ubuntu:~$ _
```

For interface iface1(192.168.2.2) of VM3

```
Ubuntu1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:89:31:d2 brd ff:ff:ff:ff:ff:ff
inet 10.0.3.15/24 brd 10.0.3.255 scope global enp0s8
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe89:31d2/64 scope link
valid_lft forever preferred_lft forever
3: enp0s9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
link/ether 08:00:27:c4:c4:2d brd ff:ff:ff:ff:ff:ff
vibhor@ubuntu:~$ sudo ip addr add 192.168.2.2/24 dev enp0s9
[sudo] password for vibhor:
vibhor@ubuntu:~$ sudo ip link set dev enp0s9 up
vibhor@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:89:31:d2 brd ff:ff:ff:ff:ff:ff
inet 10.0.3.15/24 brd 10.0.3.255 scope global enp0s8
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe89:31d2/64 scope link
valid_lft forever preferred_lft forever
3: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:c4:c4:2d brd ff:ff:ff:ff:ff:ff
inet 192.168.2.2/24 scope global enp0s9
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fec4:c42d/64 scope link
valid_lft forever preferred_lft forever
vibhor@ubuntu:~$ _
```

Set up IPv4 Forwarding on VM2



Ubuntu2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

```
vibhor@ubuntu:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
vibhor@ubuntu:~$ vi run.sh
```

Right Ctrl

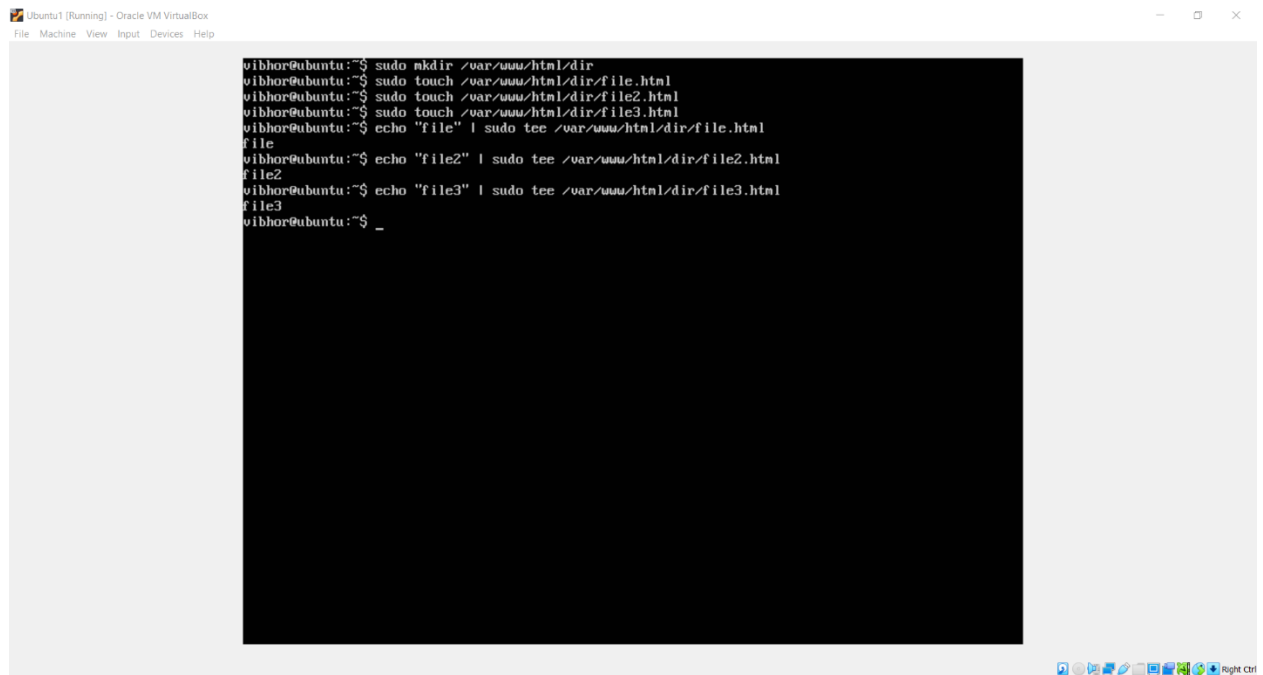
Install Apache web server on VM3

```
Ubuntu1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vibhor@ubuntu:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu xenial-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu xenial InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu xenial-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu xenial-backports InRelease
Reading package lists... Done
vibhor@ubuntu:~$ sudo apt-get install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap liblua5.1-0 ssl-cert
Suggested packages:
  www-browser apache2-doc apache2-suexec-pristine | apache2-suexec-custom openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap liblua5.1-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 11 not upgraded.
Need to get 1,559 kB of archives.
After this operation, 6,448 kB of additional disk space will be used.
Do you want to continue? [Y/n] y

Setting up libaprutil1-dbd-sqlite3:amd64 (1.5.4-1build1) ...
Setting up libaprutil1-ldap:amd64 (1.5.4-1build1) ...
Setting up liblua5.1-0:amd64 (5.1.5-8ubuntu1) ...
Setting up apache2-bin (2.4.18-2ubuntu3.17) ...
Setting up apache2-utils (2.4.18-2ubuntu3.17) ...
Setting up apache2-data (2.4.18-2ubuntu3.17) ...
Setting up apache2 (2.4.18-2ubuntu3.17) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Setting up ssl-cert (1.0.37) ...
Processing triggers for libc-bin (2.23-0ubuntu11.3) ...
Processing triggers for systemd (229-4ubuntu21.31) ...
Processing triggers for ureadahead (0.100.0-19.1) ...
Processing triggers for ufw (0.35-0ubuntu2) ...
vibhor@ubuntu:~$
```

Create web directory and adding some files to it

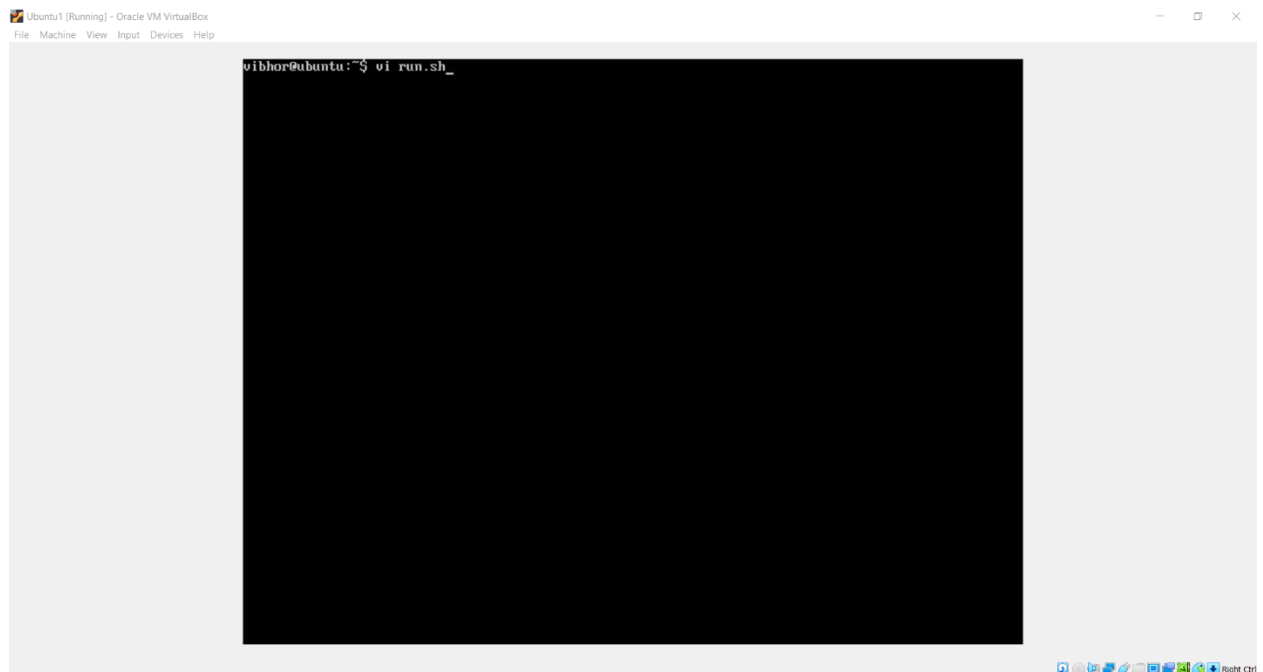


The screenshot shows a terminal window titled "Ubuntu1 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
vibhor@ubuntu:~$ sudo mkdir /var/www/html/dir
vibhor@ubuntu:~$ sudo touch /var/www/html/dir/file.html
vibhor@ubuntu:~$ sudo touch /var/www/html/dir/file2.html
vibhor@ubuntu:~$ sudo touch /var/www/html/dir/file3.html
vibhor@ubuntu:~$ echo "file" | sudo tee /var/www/html/dir/file.html
file
vibhor@ubuntu:~$ echo "file2" | sudo tee /var/www/html/dir/file2.html
file2
vibhor@ubuntu:~$ echo "file3" | sudo tee /var/www/html/dir/file3.html
file3
vibhor@ubuntu:~$ _
```

The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom status bar shows various icons and the text "Right Ctrl".

Configure bi-directional NAT and firewall on VM2



The screenshot shows a terminal window titled "Ubuntu1 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
vibhor@ubuntu:~$ vi run.sh _
```

The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom status bar shows various icons and the text "Right Ctrl".

```
#! /bin/bash -x
```

```
sudo iptables -t nat -A POSTROUTING -o enp0s10 -j MASQUERADE
sudo iptables -t nat -A PREROUTING -i enp0s10 -j DNAT --to-destination 192.168.2.2
sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -p tcp --dport 80 -j ACCEPT
sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -p tcp --dport 443 -j ACCEPT
sudo iptables -A FORWARD -s 192.168.1.1 -d 192.168.2.2 -p tcp --dport 80 -j ACCEPT
sudo iptables -A FORWARD -s 192.168.1.1 -d 192.168.2.2 -p tcp --dport 443 -j ACCEPT
sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -j DROP
```

"run.sh" 9L, 531C written

9,53

All

```
vibhor@ubuntu:~$ ./run.sh
```

```
+ sudo iptables -t nat -A POSTROUTING -o enp0s10 -j MASQUERADE
[sudo] password for vibhor:
+ sudo iptables -t nat -A PREROUTING -i enp0s10 -j DNAT --to-destination 192.168.2.2
+ sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -p tcp --dport 80 -j ACCEPT
+ sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -p tcp --dport 443 -j ACCEPT
+ sudo iptables -A FORWARD -s 192.168.1.1 -d 192.168.2.2 -p tcp --dport 80 -j ACCEPT
+ sudo iptables -A FORWARD -s 192.168.1.1 -d 192.168.2.2 -p tcp --dport 443 -j ACCEPT
+ sudo iptables -A FORWARD -i enp0s9 -o enp0s10 -j DROP
vibhor@ubuntu:~$
```

pcap output on VM1

```
Ubuntu3 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vibhor@ubuntu:~$ sudo tcpdump -r vm1.pcap
reading from file vm1.pcap, link-type EN10MB (Ethernet)
23:00:39.426728 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 1, length 64
23:00:39.428025 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 1, length 64
23:00:40.429729 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 2, length 64
23:00:40.430954 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 2, length 64
23:00:41.431507 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 3, length 64
23:00:41.432540 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 3, length 64
23:00:42.433240 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 4, length 64
23:00:42.434502 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 4, length 64
23:00:43.435055 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 5, length 64
23:00:43.436647 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 5, length 64
23:00:44.436496 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 6, length 64
23:00:44.437538 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 6, length 64
23:00:44.442832 ARP, Request who-has 192.168.1.1 tell 192.168.1.2, length 46
23:00:44.442842 ARP, Reply 192.168.1.1 is-at 08:00:27:58:6c:05 (oui Unknown), length 28
23:00:45.438473 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 7, length 64
23:00:45.439985 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 7, length 64
23:00:46.440811 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 8, length 64
23:00:46.441632 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 8, length 64
23:00:47.441052 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 9, length 64
23:00:47.442110 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 9, length 64
23:00:48.443400 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 10, length 64
23:00:48.444733 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 10, length 64
23:00:49.444604 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 11, length 64
23:00:49.445755 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 11, length 64
23:00:50.447628 IP 192.168.1.1 > 192.168.1.2: ICMP echo request, id 1190, seq 12, length 64
23:00:50.448894 IP 192.168.1.2 > 192.168.1.1: ICMP echo reply, id 1190, seq 12, length 64
vibhor@ubuntu:~$
```

pcap output on VM3

```
Ubuntu1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

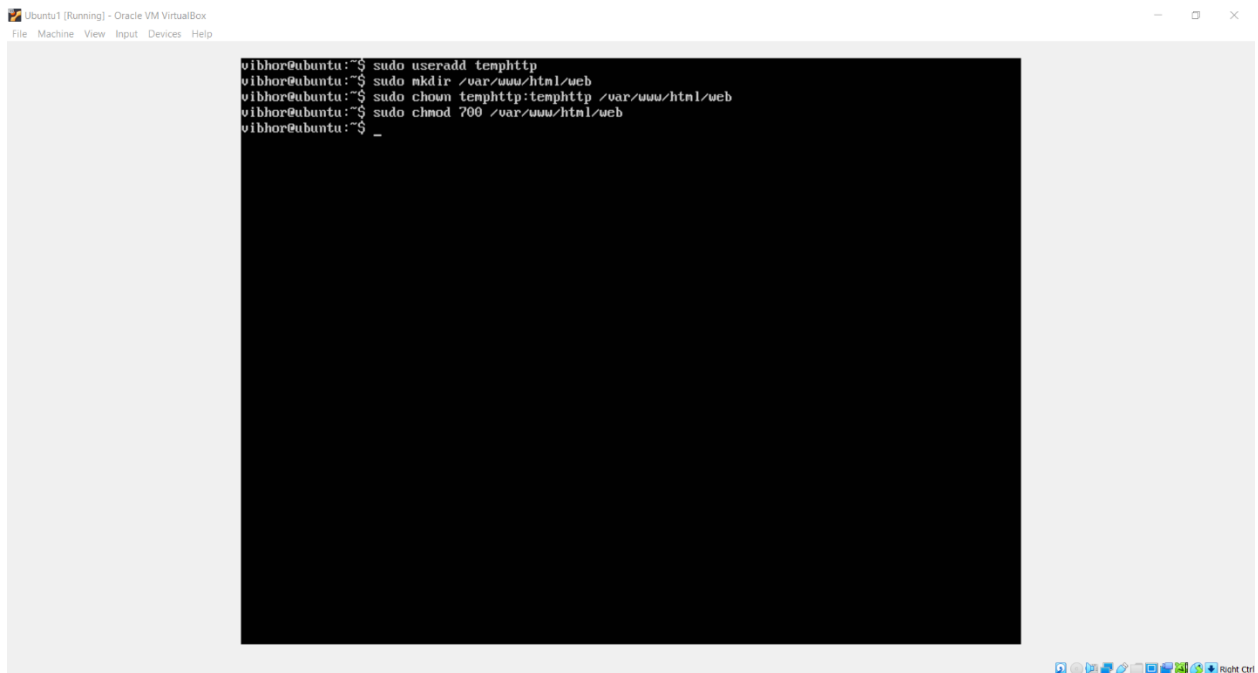
vibhor@ubuntu:~$ sudo tcpdump -r vm3.pcap
reading from file vm3.pcap, link-type EN10MB (Ethernet)
23:11:23.883806 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 35, length 64
23:11:23.883824 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 35, length 64
23:11:24.884783 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 36, length 64
23:11:24.884804 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 36, length 64
23:11:25.886786 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 37, length 64
23:11:25.886806 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 37, length 64
23:11:26.888318 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 38, length 64
23:11:26.888353 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 38, length 64
23:11:27.890503 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 39, length 64
23:11:27.890526 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 39, length 64
23:11:28.892660 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 40, length 64
23:11:28.892690 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 40, length 64
23:11:29.894225 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 41, length 64
23:11:29.894275 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 41, length 64
23:11:30.896684 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 42, length 64
23:11:30.896708 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 42, length 64
23:11:31.899326 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 43, length 64
23:11:31.899363 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 43, length 64
23:11:32.900853 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 44, length 64
23:11:32.900876 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 44, length 64
23:11:33.902944 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 45, length 64
23:11:33.902944 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 45, length 64
23:11:34.904153 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 46, length 64
23:11:34.904176 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 46, length 64
23:11:35.904946 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 47, length 64
23:11:35.904968 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 47, length 64
23:11:36.907332 IP 192.168.2.1 > 192.168.2.2: ICMP echo request, id 1379, seq 48, length 64
23:11:36.907362 IP 192.168.2.2 > 192.168.2.1: ICMP echo reply, id 1379, seq 48, length 64
vibhor@ubuntu:~$
```

Task 2

We cannot access the web server directory even with the 'setuid' bit set. The 'setuid' bit allows permission to execute an executable with the permissions of the executable owner. However, it does not provide the web server process permissions to access web directory owned by other users (in this case temphttp).

Linux ACLs allow read access to web directory by other users when rule for read access is added using setfacl.

Create user temphttp and change owner and permissions of web directory



The screenshot shows a terminal window titled 'Ubuntu1 [Running] - Oracle VM VirtualBox'. The terminal output is as follows:

```
vibhor@ubuntu:~$ sudo useradd temphttp
vibhor@ubuntu:~$ sudo mkdir /var/www/html/web
vibhor@ubuntu:~$ sudo chown temphttp:temphttp /var/www/html/web
vibhor@ubuntu:~$ sudo chmod 700 /var/www/html/web
vibhor@ubuntu:~$ _
```

The terminal window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The bottom status bar shows system icons and the text 'Right Ctrl'.

Set 'setuid' bit of web server process and try to write to web directory (unsuccessful access)

```
Ubuntu1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vibhor@ubuntu:~$ sudo chmod u+s /usr/sbin/apache2
[sudo] password for vibhor:
vibhor@ubuntu:~$ ls -l /usr/sbin/apache2
-rwsr-xr-x 1 root root 662560 Aug 13 2020 /usr/sbin/apache2
vibhor@ubuntu:~$ cd /var/www/html/web
-bash: cd: /var/www/html/web: Permission denied
vibhor@ubuntu:~$
```

Specific read access to web server process using Linux ACLs (successful access)

```
Ubuntu1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vibhor@ubuntu:~$ sudo setfacl -m u:vibhor:r /var/www/html/web
vibhor@ubuntu:~$ getfacl /var/www/html/web
getfacl: Removing leading '/' from absolute path names
# file: var/www/html/web
# owner: tenhttp
# group: tenhttp
user::rwx
user:vibhor:r--
group::---
mask:r--
other::---

vibhor@ubuntu:~$ ls /var/www/html/web
vibhor@ubuntu:~$
```